Asyncio Plugin

Asynchronous AI Chat for Limnoria IRC

version 1.1 | python 3.9+ | limnoria compatible | OpenAI powered | license MIT

async safe | context isolated | production ready

Asyncio is a production-ready AI chat plugin for Limnoria IRC bots. It provides natural conversation with OpenAI models while ensuring safe multi-user behaviour, isolated memory, and stable long-running operation.

Bot Name: Puss
Plugin: Asyncio — Asynchronous AI Chat for Limnoria IRC
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1 Commands Cheat Sheet

Command	Example Usage	Description

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💬 `!chat <message>`	`!chat What is the acceleration due to gravity on Earth?`	Sends your message to the AI assistant. Moderation, cooldowns, and token limits apply automatically.
🧹 `!reset`	`!reset`	Clears your conversation memory for the current channel or private message only.

💡 **Tip:** Each channel and private message keeps its own conversation context.

2️⃣ Conversation Memory

The plugin maintains conversation history per **user AND channel**.

This ensures:

- Conversations in different channels stay separate
- Private messages are treated as their own context
- Users do not share memory with each other
- History auto-trims to prevent growth

Example

In `#test`: !chat Remember my favourite colour is blue !chat What is my favourite colour?

Bot remembers.

In another channel: !chat What is my favourite colour?

Bot does **not** remember.

3️⃣ Plugin Configuration (`@config`)

Setting	Example	Description
🔢 `maxUserTokens`	`@config plugins.Asyncio.maxUserTokens 2048`	Maximum tokens allowed per user input.
⏱️ `cooldownSeconds`	`@config plugins.Asyncio.cooldownSeconds 5`	Minimum delay between messages per user & channel.
🐞 `debugMode`	`@config plugins.Asyncio.debugMode True`	Enables detailed logging for troubleshooting.

Setting	Example	Description
🤖 `botnick`	`@config plugins.Asyncio.botnick Puss`	Bot name used in responses.
🌐 `language`	`@config plugins.Asyncio.language British`	English style preference for AI responses.

4 Best Practices

☑ Keep Messages Reasonably Concise

- Stay under `maxUserTokens`.
- Split very long content into multiple messages.

☑ Moderation-Friendly Input

- Avoid offensive, illegal, or harmful content.
- If a message is flagged, rephrase it.

☑ Understand Cooldowns

- Cooldowns apply per user AND per channel.
- Other users are unaffected.
- Other channels are unaffected.
- Private messages have their own cooldown.

☑ Reset When Needed

- Use `!reset` to start a fresh conversation.
- Only resets the current channel or PM.

☑ Debugging & Troubleshooting

- Enable `debugMode` for full logs.
- Check Limnoria logs if AI stops responding.

☑ Personalisation

- Adjust `botnick` and `language` to change style.

5 Optional Tips

- Encourage users to ask clear, structured questions.
- Avoid sending many queries rapidly (may hit API limits).
- Keep dependencies (`openai`, `dotenv`, etc.) updated.

- Restarting Limnoria clears all conversation memory.

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6 Operator Notes

This plugin is designed for production IRC use.

It includes:

• Async-safe execution (non-blocking bot behaviour)
• Built-in moderation filtering
• Token length protection
• Per-user + per-channel memory isolation
• Automatic history trimming
• Safe cooldown enforcement

The design prevents:

• Cross-channel memory leakage
• One user affecting another's context
• Excessive API usage
• Runaway memory growth

Optional future upgrades include:

• Persistent memory to disk
• Admin control commands
• Usage statistics
• Channel enable/disable toggles

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7 Exporting to PDF

⚡ Quick Command (Pandoc)

```
pandoc AsyncioPluginGuide.md -o AsyncioPluginGuide.pdf \
    --pdf-engine=xelatex \
    -V geometry:margin=1in

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```